

ITU UMC DATA HACKATHON 2026

Universal & Meaningful Connectivity in a Hyper-Connected City

Development Integrated Framework

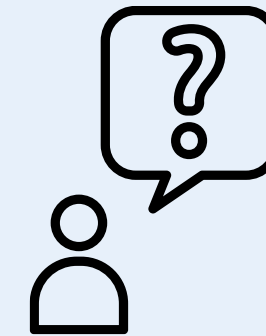
TEAM High-Five

Problem Statement

Even where the city looks hyper-connected, are there hidden gaps in digital connectivity?



How can problems be identified?

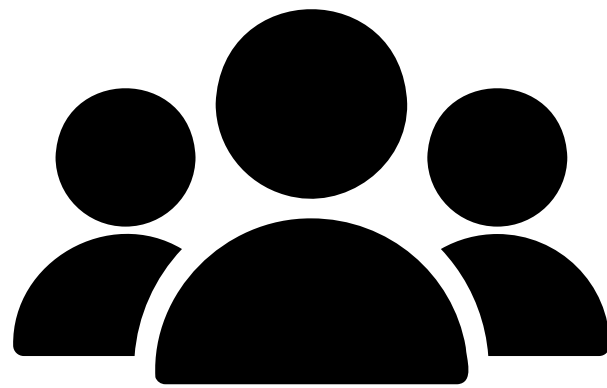


How can it be measured?



How can it be supported?

Why is this important?



People?

Who is underserved?



Place?

Where are the gaps?

Even where UMC has been achieved, some populations may remain systematically underserved

What We Do?

This project applies the Universal Meaningful Connectivity (UMC) framework to Seoul, **identifies digital deserts at the policy implementation level**, and **specifies vulnerable populations** affected by digital exclusion.

01

Apply

Apply UMC framework
to Seoul context

02

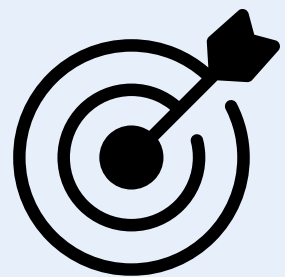
Identify

Identify digital deserts
at policy level

03

Develop

Develop integrated
framework for action



Broader applicability, plus identification of specific problems

What We Use?

Data Source

Analysis 1 & 2

Dataset name	Source	Link	Coverage	Key variables	Format	Access method
Seoul Digital Competency Survey	Seoul AI Foundation	https://saif.or.kr/dataroom/2564	Geographic: Seoul Time: 2023 used	devices, skills, safety; survey weights	CSV/XLSX	data provision upon official request
Cellular Base Station Data	Spectrum Resource Mgmt System	https://spectrummap.kr/gis/mobile_service.do?menuNo=300480	Geographic: Seoul Time: 2023, 2024	4G/5G cell-site locations		Public Open Data
NIA Communication Quality (CQ)	Korean Public Data Portal	https://www.data.go.kr/data/15120591/fileData.do	Geographic: Seoul Time: 2023, 2024	district-level download speed		
Public Free Wi-Fi Installation Data	Local Admin WiFi License Data	https://www.localdata.go.kr/lif/lifeCtacDataView.do	Geographic: Seoul Time: 2023, 2024	public Wi-Fi installation locations		
Telecommunication Indicators	SKT Telecom Data	https://www.data.go.kr/data/15127168/fileData.do#	Geographic: Seoul Time: 2023, 2024	mobile data use; online service days; fee delinquency rate		
Average Monthly Household Income by District	Seoul Commercial Analysis Service	https://data.seoul.go.kr/dataList/OA-22167/S/1/datasetView.do	Geographic: Seoul Time: 2023, 2024	average monthly household income		
Seoul Survey	Seoul Open Data	https://data.seoul.go.kr/dataList/OA-15564/F/1/datasetView.do	Geographic: Seoul Time: 2023, 2024	digital service use; demographics, income, housing, district/year		

Analysis 3

Dataset name	Source	Link	Coverage	Key variables	Format	Access method
Daangn Community Post	Daangn Market	daangn.com/kr/community/s	Geographic: Seoul dong fields Time: 2024-2026 crawl period	post text; gu/location; createdAt; subject/category	CSV/XLSX	Web Scraping
Seoul Living/Floating Population	Seoul Open Data	data.seoul.go.kr OA-14991	Geographic: Seoul Time: 2023-2026 used as available	hourly/dong population; daytime population; age shares; dong/district code		Public Open Data

01

01

Development-Integrated Framework

Theoretical foundation and analytical design

Considerations

Theoretical Considerations

Evidence-based policy must confront problems that exist yet stay hidden from conventional evidence



Data



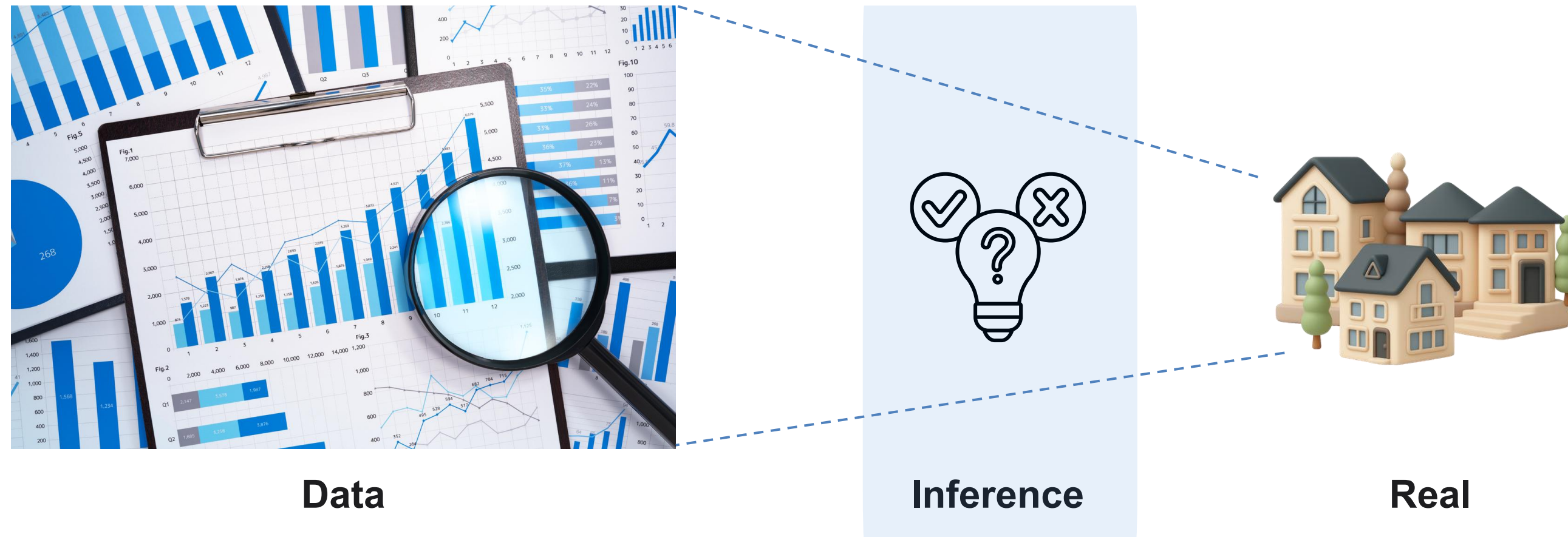
Real

The problems of hyper-connected cities exist independently of their observation in data

Considerations

Theoretical Considerations

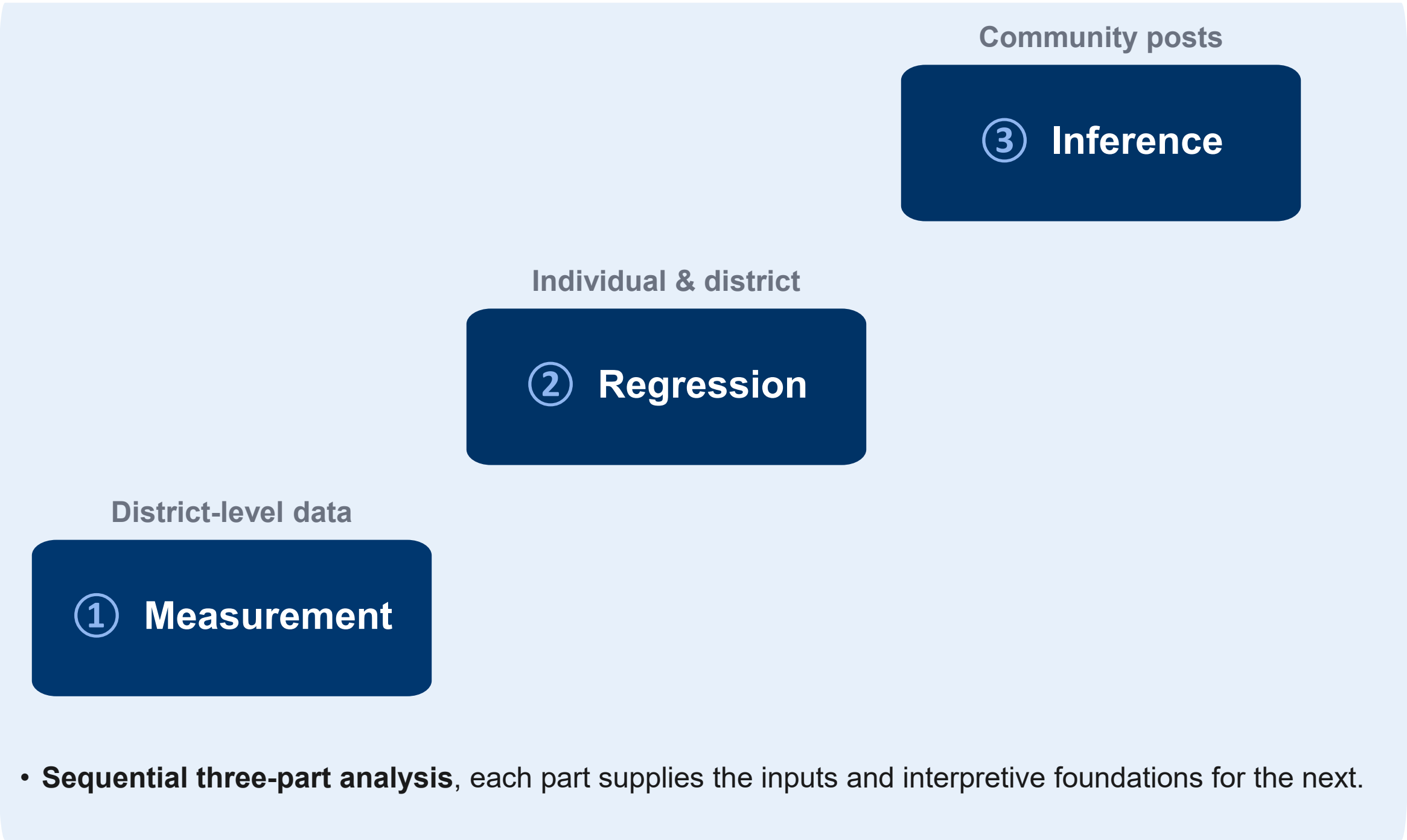
Separating ontology from epistemology lets us **infer underlying real-world mechanisms from empirical observation**



The approach **transfers across contexts** by making explicit which elements **must be empirically verified** when inferring latent conditions from data

Considerations

Metatheoretical framework linking the three analyses under one structural logic



(1) Measurement Using a Composite Index

- Six UMC dimensions measured across Seoul's 25 districts

(2) Regression-Based Analysis

- HLM of individual and district characteristics

(3) Bayesian & Multi-Agent Analysis

- Inference of latent conditions from community-post evidence

02

02

ETL Pipeline & Analytical Method

Data engineering and the three analytical layers

Two Quantitative Analyses

Measure where disparities sit (3-1) and for whom they appear (3-2)

ANALYSIS 3-1 Composite Index

UNIT
Seoul's 25 districts, the unit of policy implementation.

METHOD
Within-Seoul Min-Max scaling per indicator, then an equal-weighted mean of six dimensions to give score_UMC (2023 and 2024).

WHY MIN-MAX
Seoul already passes most basic UMC thresholds, so the question becomes relative position, not presence or absence.

KEYWORDS
six UMC dimensions · within-Seoul scaling · equal weights

3-1 locates the gap by place; 3-2 explains it by person. Together they set up the platform-signal layer in 3-3.

ANALYSIS 3-2 Multilevel Model (HLM)

UNIT
10,000 individuals nested in 25 districts (2023-2024 pooled).

OUTCOME
Digital-usage score: eight Seoul-Survey items, 0 to 80.

PREDICTORS
Level 1 individual (education, age, household, income, occupation, housing) and Level 2 district (Connectivity, Available for Use, Affordability).

MODELS
Unconditional (ICC) to individual to + district to + interactions to 65+ threshold.

AIM
Multilevel association, not causal inference.

Why Danggeun?

A hyperlocal platform whose posts surface lived-experience signals that surveys and admin data miss.

ANALYSIS UNIT

Individual neighborhood-life post.

SOURCE

Danggeun, a Korean local community platform.

WINDOW

May 2024 to March 2026, across Seoul's 25 districts.

CAVEAT

Not a representative sample. It observes only users who chose to post (selection bias).

Geo-tagged

Every post carries dong and district metadata, so posts link directly to the 3-1 UMC index.

Citywide

Users are spread across all 25 districts, which makes district-level aggregation feasible.

Lived events

Posts record problems met in daily life, surfacing signals that surveys and administrative data miss.

Treat platform signals as a lower bound and experientially selective evidence, not population prevalence.

Analysis 3-3 · ETL Pipeline

All community-post IDs in the range 727,174,000 – 733,968,953 (≈ 6.79M) were requested sequentially through Danggeun's structured JSON route and filtered to Seoul

HTTP RESPONSE HANDLING

Code	Meaning	Action
200	Active post	Parse and store
410	Deleted (metadata retained)	Store with status flag
404	Empty ID	Skip
429	Rate limited	Reduce request rate

410 responses are retained with a status flag, which enables the accessibility analysis on the following slides.

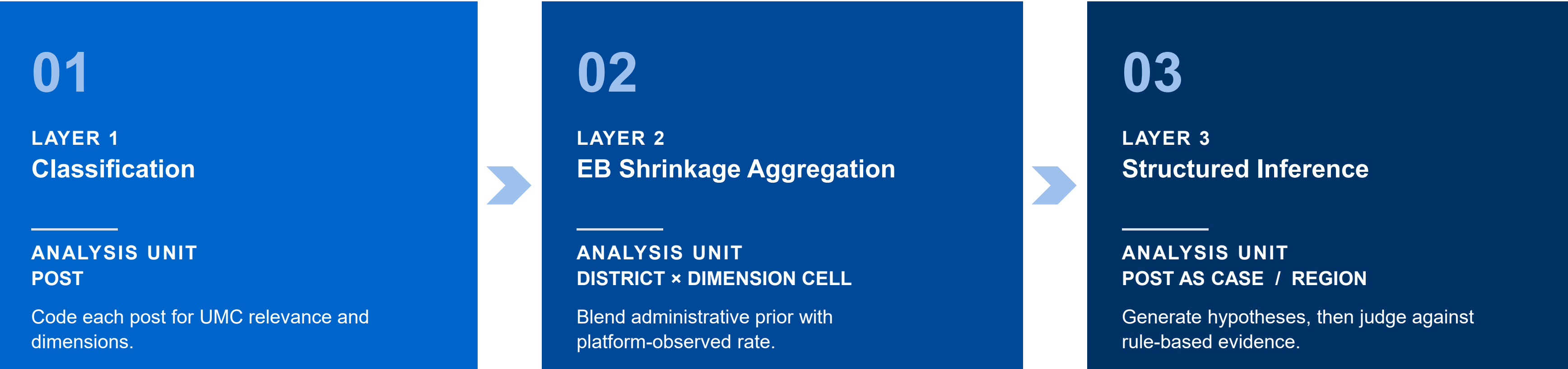
RATE CONTROL & OPERATION

Parameter	Setting
Engine	Asynchronous (aiohttp)
Concurrency / batch	10 concurrent · 200 per batch
Request rate	Start 15 · adaptive 2–30 req/s
Back-off (HTTP 429)	Rate ×0.5 · 10s cooldown
Recovery	+10% after 5 stable batches
Partitioning	2,000-ID chunks (~3,400 total)
Resume	Checkpoint every 500 rows

Analysis 3-3, Platform-Visible Signal Analysis

A three-layer pipeline that turns everyday community-platform text into auditable, UMC-coded regional signals.

Dimension labels follow each analysis: Analysis 3-1 uses **Connectivity / Available for Use**, read here as the equivalent **Connection Quality / Availability for Use**.



Unit of analysis shifts deliberately: **post** → **district × dimension cell** → **post-as-case / region**

Relevance & Dimension Classification

Turn unstructured everyday Korean text into theory-coded UMC signals.

ANALYSIS UNIT

Individual post (Neighborhood-Life community post).

GOAL

Decide UMC relevance (Y / ? / N) and assign one or more of the six UMC dimensions to every relevant post.

PURPOSE

Build a clean candidate signal base that captures lived-experience deficits administrative and survey data miss.

KEYWORDS

LLM structured coding · seed list + codebook · multi-label · few-shot · language-independent

- 01 **Crawl & clean**
All Seoul Neighborhood-Life posts, 25 districts, May 2024 – Mar 2026. Remove inaccessible, duplicate, and advertising posts.
- 02 **Seed-keyword filter**
Apply the UMC six-dimension seed list and expanded dictionary to collect candidate posts (149,733 candidates).
- 03 **LLM relevance coding**
An LLM, used as a structured coding assistant on a fixed codebook, sorts candidates into bounded categories: relevant (Y) / indeterminate (?) / not relevant (N). 131,792 dong-matched; Y = 7,136.
- 04 **Dimension assignment**
Each relevant post is mapped to ≥ 1 of six dimensions: Connection Quality, Availability for Use, Affordability, Devices, Digital Skills, Safety. Multi-label: dimension totals need not sum.

Empirical-Bayes Shrinkage Rate Aggregation

Compare administrative expectations with platform-visible signals on a common rate scale.

ANALYSIS UNIT

District × dimension cell (25 districts × 6 dimensions).

GOAL

Blend the Part-1 administrative prior with the platform-observed posting rate into a precision-weighted posterior, then quantify the divergence.

PURPOSE

Locate where platform voices diverge from administrative indicators. The result is a candidate list for review, not a measure of deprivation size.

KEYWORDS

living-population exposure · precision-weighted posterior · shrinkage strength κ · z_shift

- **01 Aggregate to cells**
Classified posts are pooled into district × dimension cells (multi-label, so cell totals are not mutually exclusive).
- **02 Exposure-adjusted observed rate**
Observed rate = classified posts ÷ living-population exposure, where the denominator is authoring-time person-years. Reported as posting signals per 100,000 person-years.
- **03 Administrative prior**
The Part-1 UMC score is reversed into a deficiency orientation, then rescaled to the dimension's observed-rate range to give the administrative prior rate.
- **04 EB posterior estimate**
Precision-weighted mean of prior and observed, with prior weight $\kappa / (\text{exposure} + \kappa)$. Shrinks toward the prior when between-district variation $\approx$ sampling noise.
- **05 Shift & z_shift**
 $shift = \text{posterior} - \text{prior}$; z_shift is the within-dimension standardized shift, used to rank priority-review cells across dimensions.

Read z_shift as: the distance between administrative expectation and platform signal, not the magnitude of deprivation. Large cells flag where to look closer.

Structured Inference, Hypothesis Generation

Three isolated agents read the same post and propose what underlying problem it points to.

ANALYSIS UNIT

Individual post, re-read as a case (only $|z_shift| \geq 1.5$ cells enter).

GOAL

Generate text-grounded regional hypotheses about the latent digital-connectivity problem behind a post.

PURPOSE

Information isolation: agents never see the EB labels or each other, so hypotheses cannot be back-fitted to known results.

KEYWORDS

information isolation · text-only · blind agents · abductive · forward · sequential

A Extract

Prepare a minimal post record: text, region / time, and dimension codes only (deterministic).

B Reason ×3 (text-only, isolated)

Each agent generates hypotheses independently, blind to the others and to the EB divergence labels.

Abductive

ASKS
What hidden condition best explains this post?

Infers the underlying barrier behind the observed signal.

Forward

ASKS
If that condition held, what would we expect to see?

Predicts the signals a given condition should produce.

Sequential

ASKS
Where along access → use → resolution does it break?

Reconstructs the user's path to locate the point of failure.

Next: the three hypothesis sets pass to Layer 3, Step B, where they are specified, evidenced, and judged against registered data.

Structured Inference, Evidence & Judgment

Turn hypotheses into auditable verdicts using a fixed, rule-based evidence procedure.

ANALYSIS UNIT

Regional hypothesis × registered district evidence.

GOAL

Assign supported / refuted / undetermined, only when the registered evidence directly bears on the hypothesis.

PURPOSE

Make every verdict auditable and reproducible through a fixed evidence rule, not analyst discretion.

KEYWORDS

registered variables · rule-based evidence · auditable verdict · $|z_shift| \geq 1.5$ gate

- C

Specify data
Convert each Step-B hypothesis into a list of registered variables and explicit verification rules, before any evidence is pulled.
- D

Build evidence
Assemble a rule-based evidence bundle from registered aggregate tables and external district indicators (percentile / mean).
- E

Judge
Assign supported / refuted / undetermined. 'Supported' requires external district data plus a direct quote; Bayesian outputs are context only. Only 'supported' hypotheses are coded.

Status: an exploratory layer. It does not estimate prevalence or prove causation. The $|z_shift| \geq 1.5$ gate selects which district-dimension cells are examined at all.

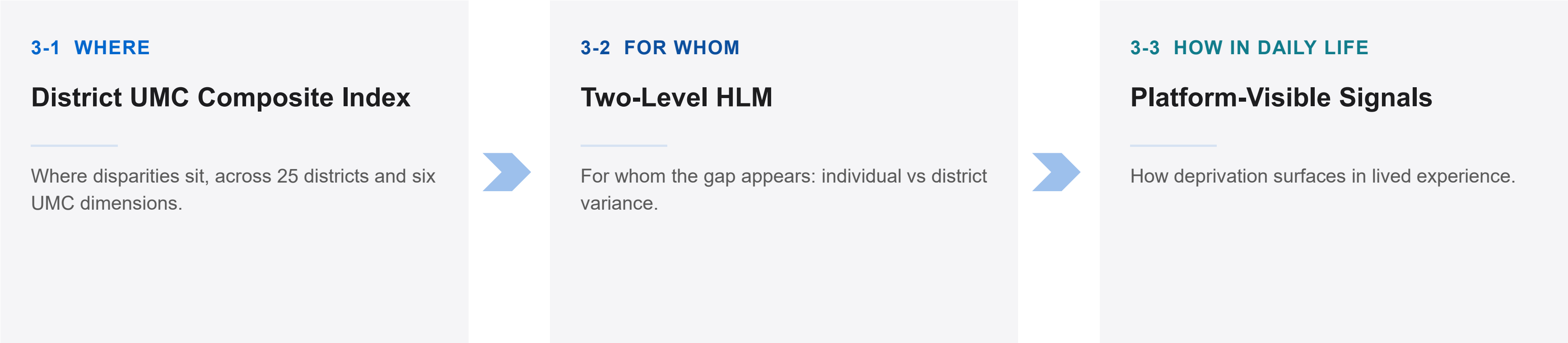
03

Project Results

Where disparities sit (3-1) → for whom (3-2) → how they are lived (3-3).

One Investigation, Three Connected Analyses

Each analysis hands its output to the next.



ALL THREE → PART 04 people-centered, place-sensitive, and a capability / safety / affordability bundle.

Indicator Composition

UMC six-dimension indicators for Seoul, 25 districts.

Dimension	Indicator	Definition
Connectivity	station_density_4g	4G/5G base-station density per 1,000 daytime residents
	station_density_5g	4G/5G base-station density per 1,000 daytime residents
	download_speed	District average download speed
Available for Use	mobile_data_usage	Monthly mobile data usage, weighted by daytime floating population
	online_service_days	Online service-use days across finance, shopping, video, delivery
	wifi_density	Registered public Wi-Fi access points
Affordability	avg_income	Average monthly household income by district
	delinq_rate	Telecom fee delinquency rate, reverse-coded

Dimension	Indicator	Definition
Devices	device_home_core	Core device types owned in the household
	device_use_core	Core device types actually used by individuals
Digital Skills	Basic	Basic skills
	Applied	Applied skills
	Advanced	Advanced skills
	Cyber activity	Cyber activity
Safety	safety_behavior	Security behavior
	safety_awareness	Security awareness

District UMC Composite Index

Within-Seoul composite score, 2023 vs 2024 (relative position, not absolute attainment).

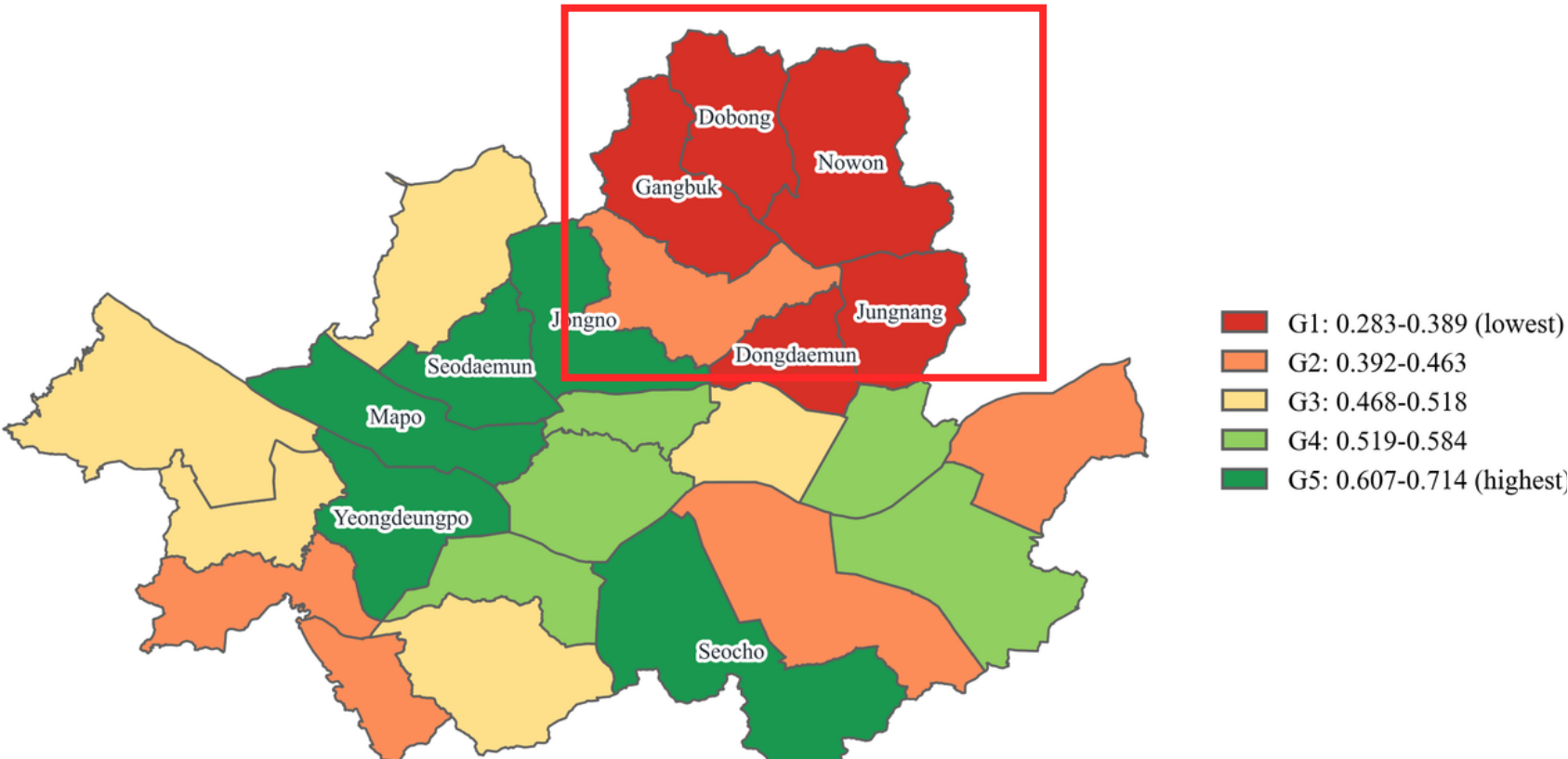


Figure. 2023 Seoul UMC Index

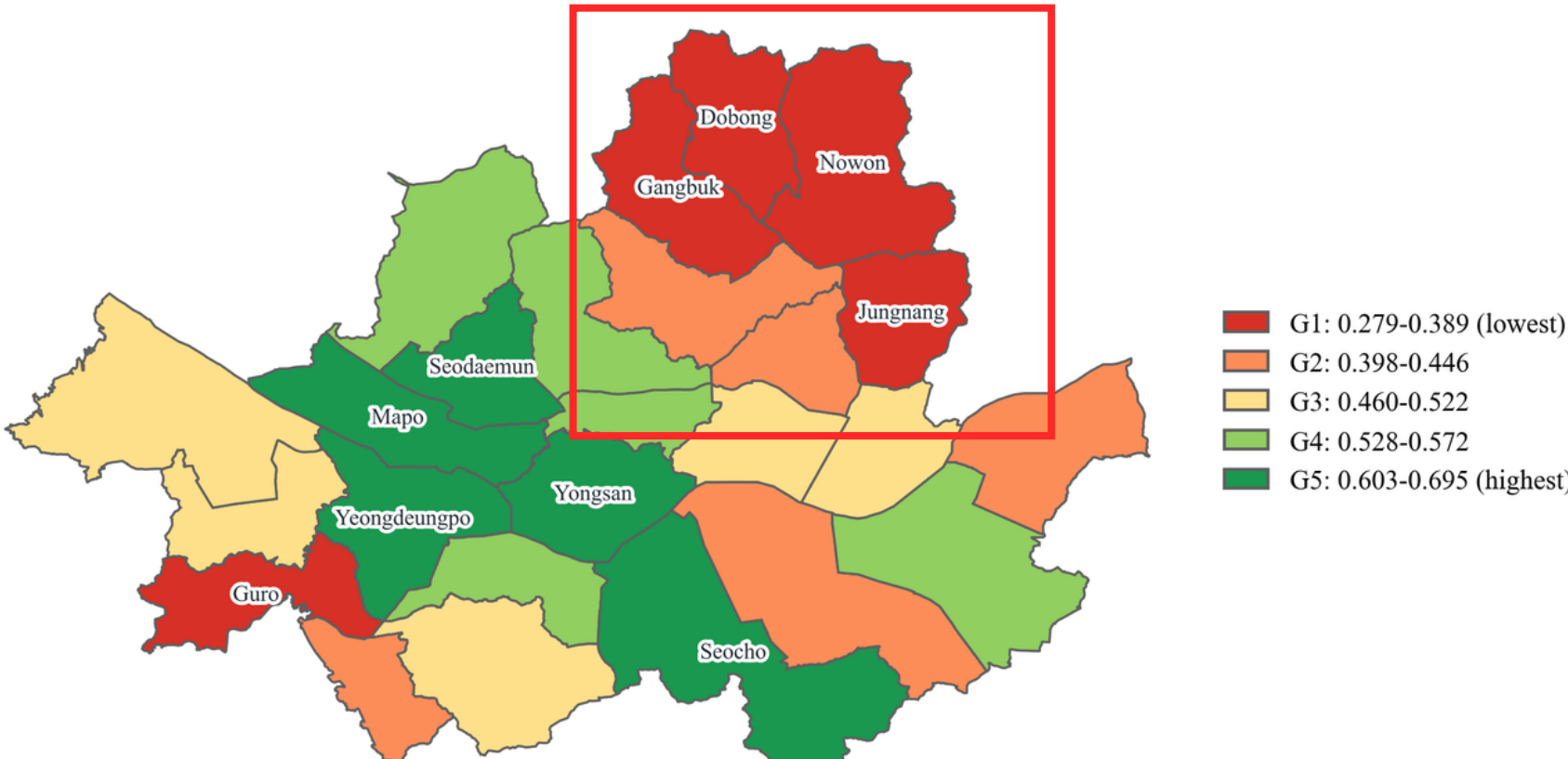


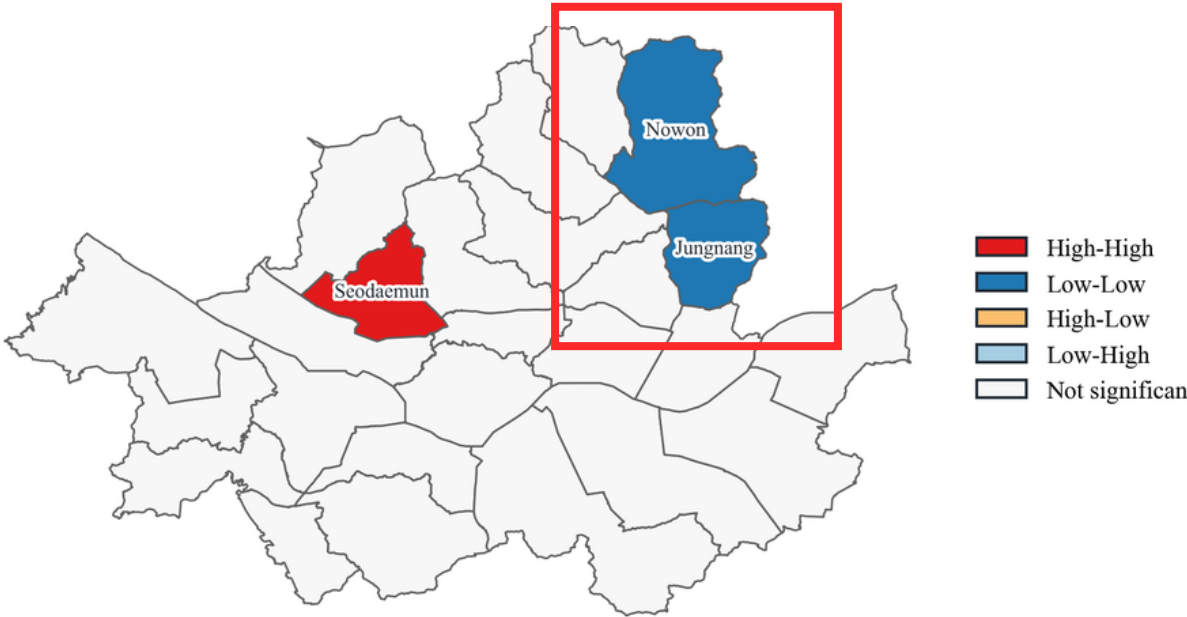
Figure. 2024 Seoul UMC Index

KEY Point The 2024 composite runs from 0.279 (Jungnang-gu) to 0.695 (Seocho-gu); the gap is dimension-specific, not infrastructure alone.

Spatial Autocorrelation (LISA)

Local clusters by UMC dimension, 2024.

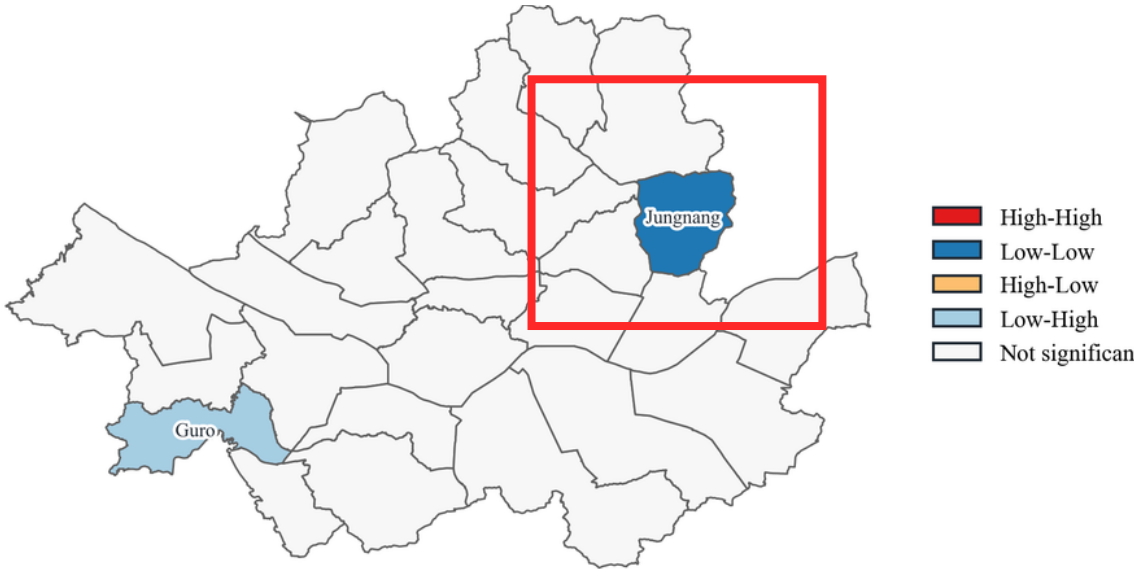
Safety



Available for Use



Digital Skills



The district × dimension UMC index now feeds 3-2 as Level-2 covariates (Connectivity, Available for Use, Affordability) and 3-3 as the Empirical-Bayes prior.

Variable Description

Level-1 individual and Level-2 district variables.

Variable	Description	Level	Scale
digital_use_score	Digital service usage score (sum of 8 items)	L1	0-80 (continuous)
age_c	Age (grand-mean centered, M = 49.3)	L1	years
elderly	Older adults aged 65 or above (Model 4 threshold)	L1	0/1
female	Female (ref: male)	L1	0/1
Edu	Education: <= middle school / high school / graduate (ref: college)	L1	0/1 each
Income	Monthly income (grand-mean centered)	L1	10K KRW
job	Job: professional / blue-collar / other (ref: white-collar)	L1	0/1 each
disabled_bin	Disability status	L1	0/1

Variable	Description	Level	Scale
hh_size_1	Single-person household	L1	0/1
housing_detached / _multi / _other	Housing: detached / multi-family / other (ref: apartment)	L1	0/1 each
year	Year 2024 dummy (ref: 2023)	L1	0/1
low_edu	Lower educational attainment (<= middle school)	L1	0/1
Connectivity	UMC connectivity/infrastructure score (normalized)	L2	0-1
Available for Use	UMC available-for-use score (normalized)	L2	0-1
Affordability	UMC affordability score (normalized)	L2	0-1

RECEIVED The 3-1 district UMC index enters here as Level-2 covariates (three infrastructure dimensions).

Descriptive Statistics

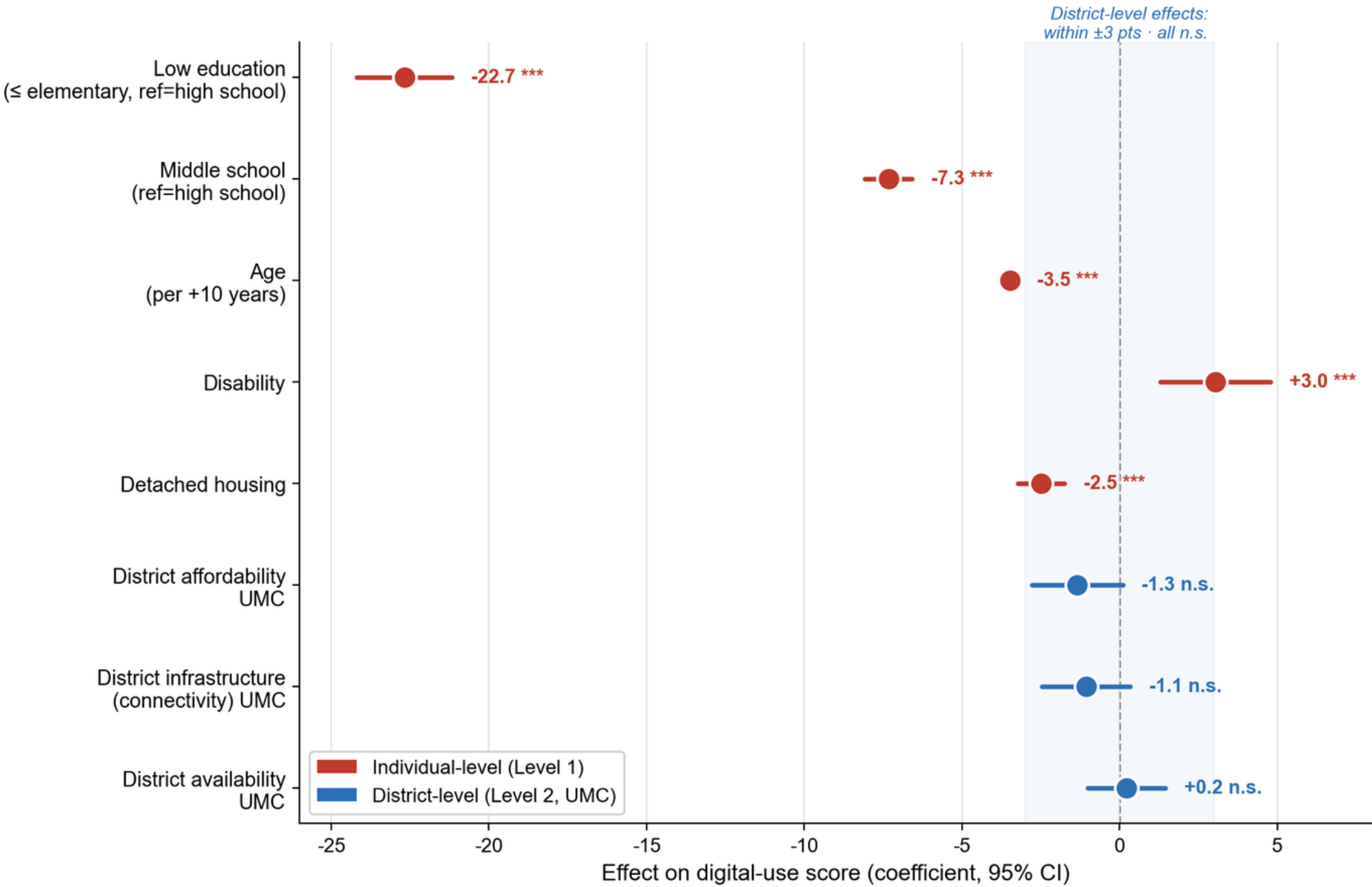
Pooled 2023-2024 Seoul Survey sample.

Variable	N	Mean	SD	Min	Max
Digital usage score	10,000	60.88	16.84	0	80
Age	10,000	49.32	16.23	15	99
Older adults (age 65+), proportion	10,000	0.15	—	0	1
Female, proportion	10,000	0.51	—	0	1
Education: <= middle school	10,000	0.07	—	0	1
Education: high school	10,000	0.25	—	0	1
Education: graduate school or higher	10,000	0.13	—	0	1
Disability status	10,000	0.03	—	0	1
Monthly income (10K KRW)	10,000	237.55	209.73	0	1,050
Single-person household	10,000	0.17	—	0	1
Cluster-level Connectivity Index (norm.)	50	0.5	0.18	0.18	0.98
Available for Use (norm.)	50	0.5	0.19	0.12	0.96
Affordability (norm.)	50	0.5	0.22	0.07	1

KEY TAKEAWAY N = 10,000 individuals nested in 25 districts (about 400 each); the mean digital-use score is 60.88 out of 80.

Individual Characteristics Dominate

Individual characteristics, not regional conditions, explain most variation in digital usage.

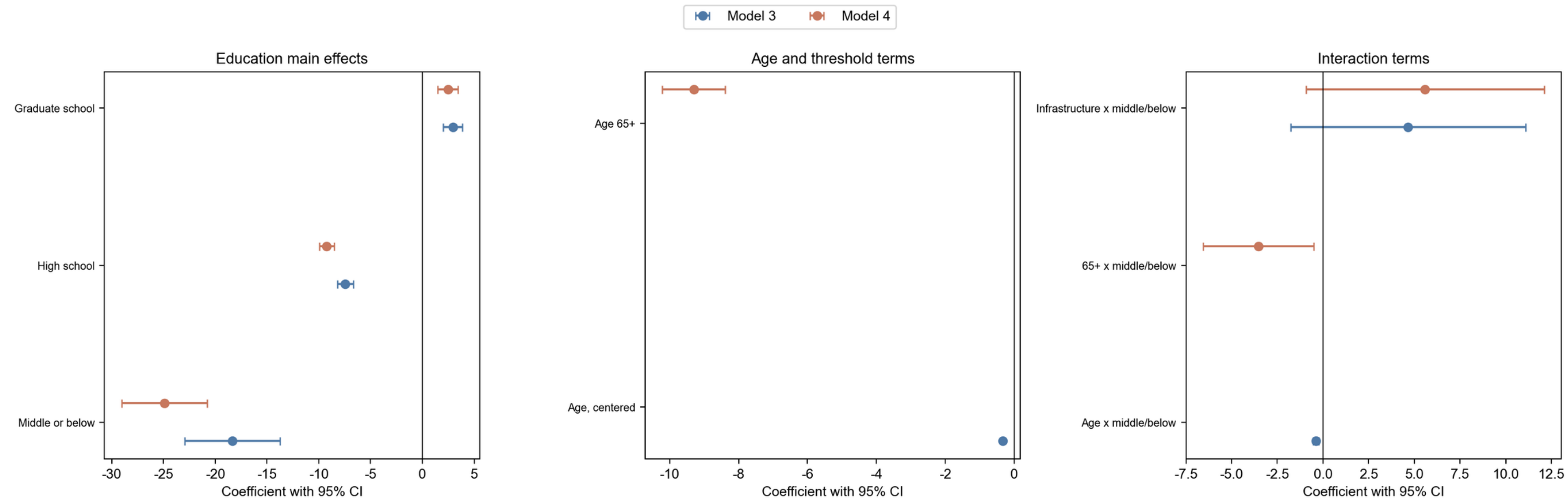


The strongest negative associations

lower educational attainment (middle school or below) and **older age**.

Deficits Concentrate at Age 65+

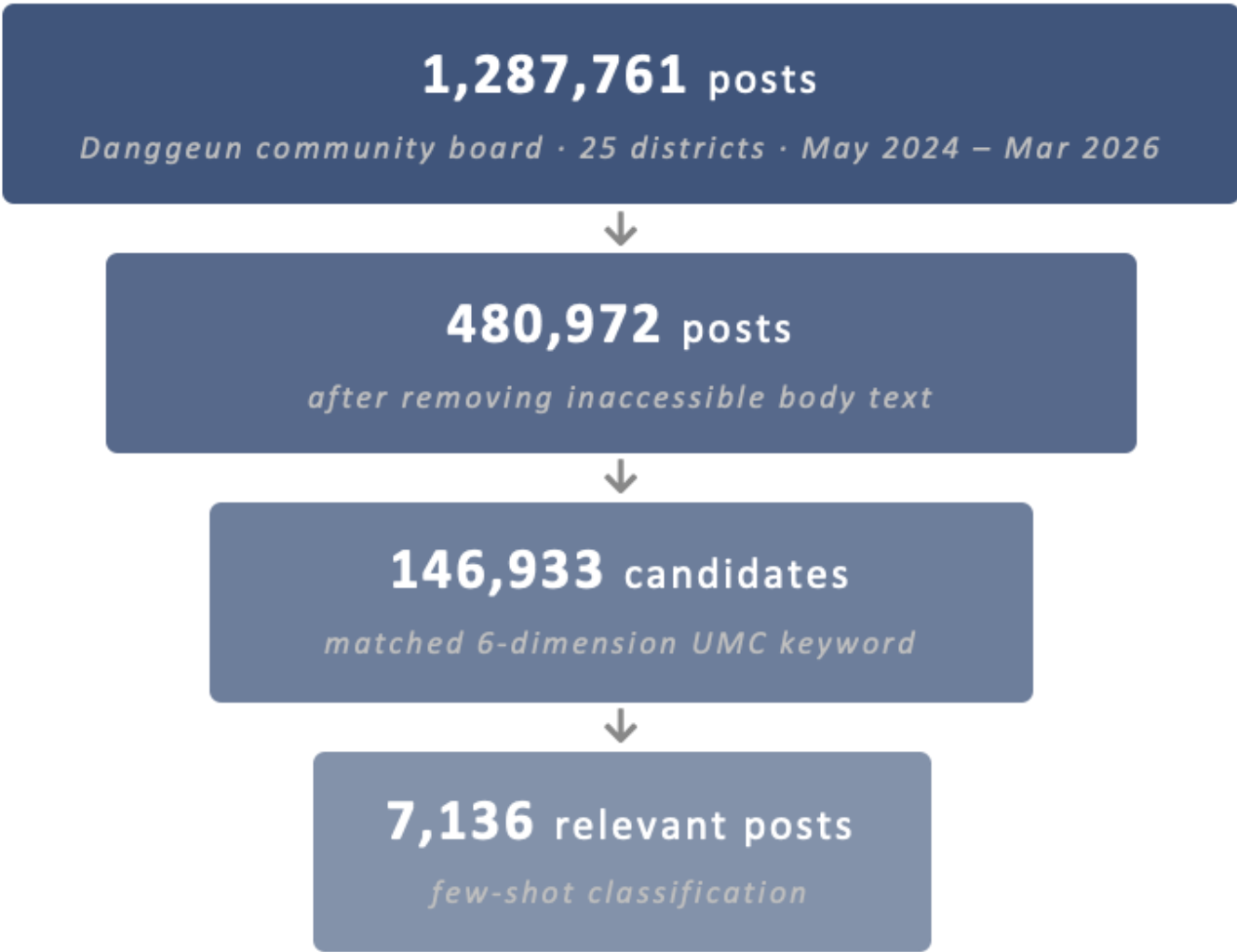
Digital-experience deficits are most pronounced among adults **aged 65 and above**, beyond the continuous age effect.



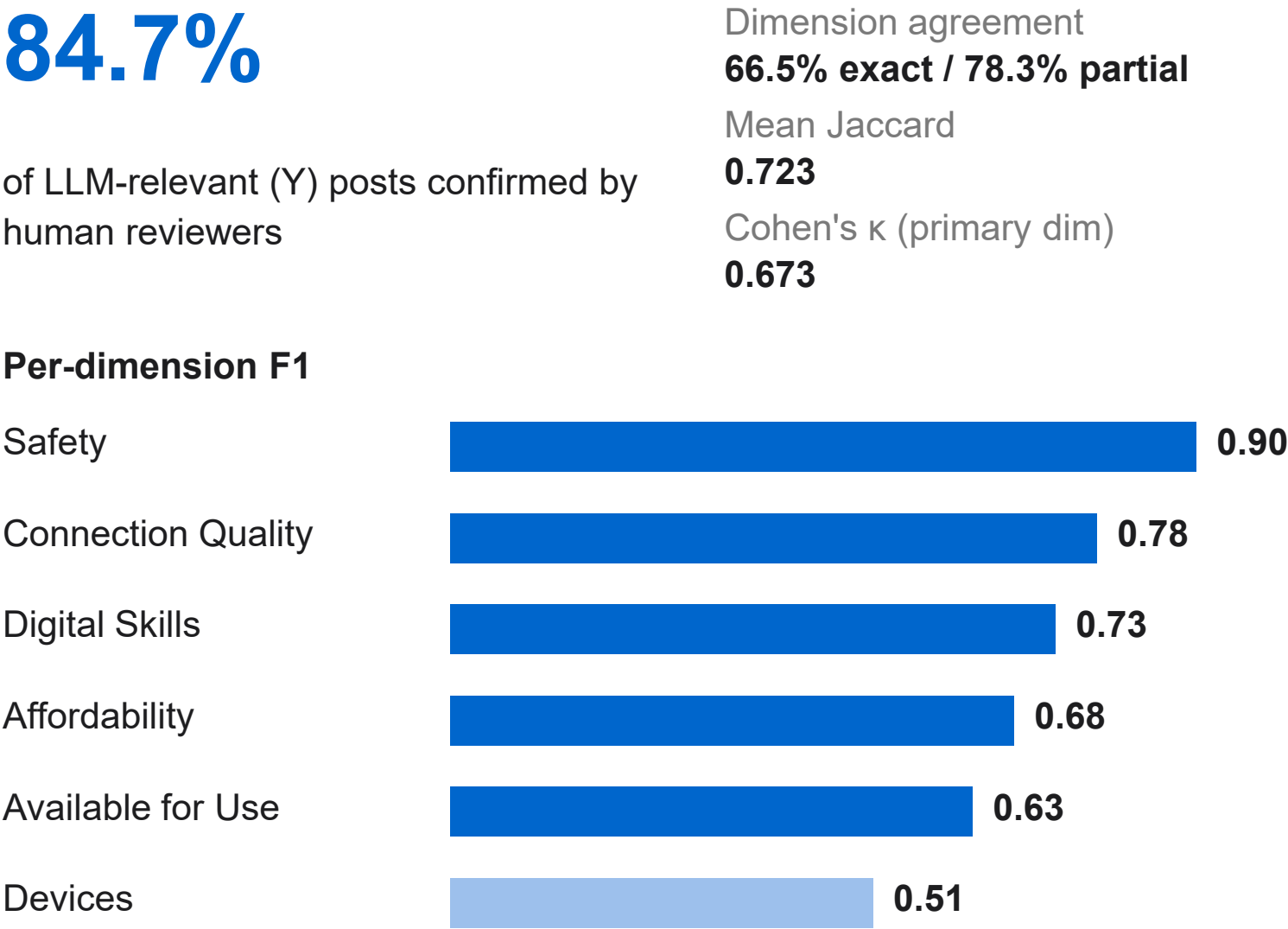
CARRIES FORWARD District context is weak; in 3-3 it serves only as an interpretive guardrail, never a direct cause of individual use.

Prior-to-Signal Divergence (EB z_shift)

ANALYSIS 3-3 · PLATFORM-VISIBLE SIGNALS



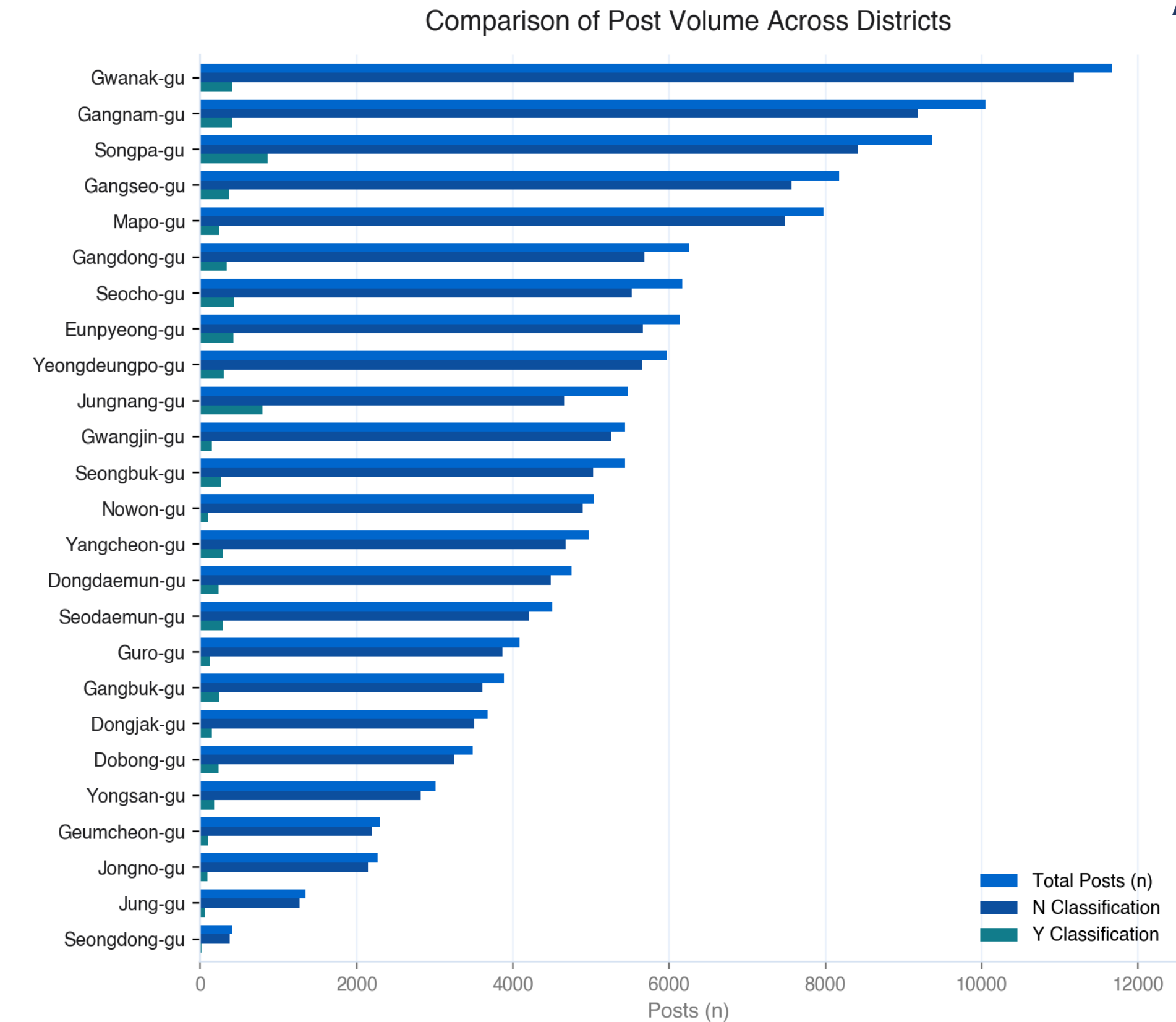
Human reviewers re-judged a review sample of LLM-coded posts (report Table 7).



RECEIVED 3-1 index = the Empirical-Bayes prior; the 3-2 result = an interpretive guardrail.

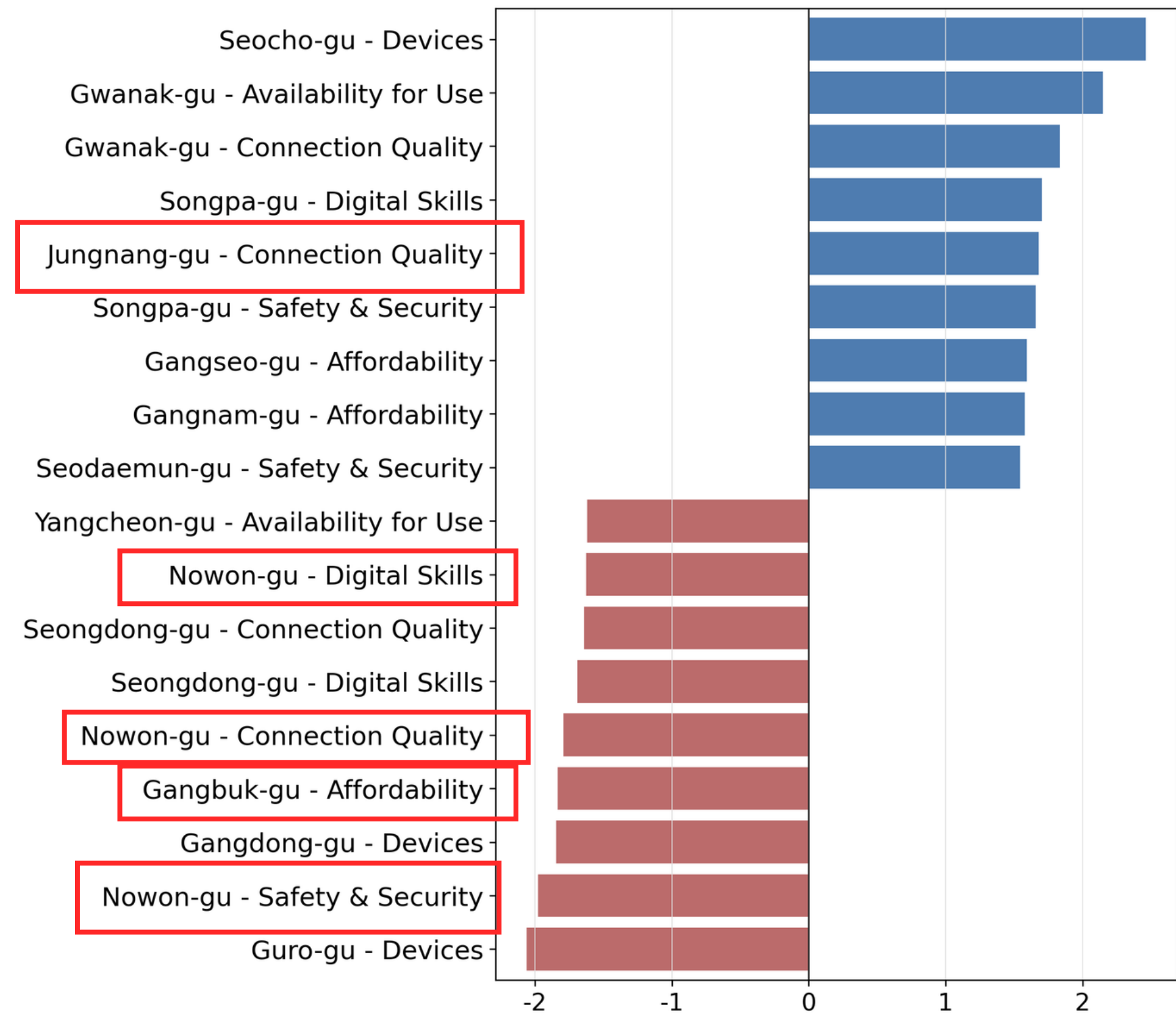
Platform Signal Volume by District

ANALYSIS 3-3 · PLATFORM SIGNAL VOLUME



- Highest total post volume: Gwanak-gu (12,000)
- Lowest total post volume: Seongdong-gu (400)
- Highest number of Y-classified posts: Songpa-gu (8,408)
- Highest proportion of Y-classified posts: Jungnang-gu (14%)

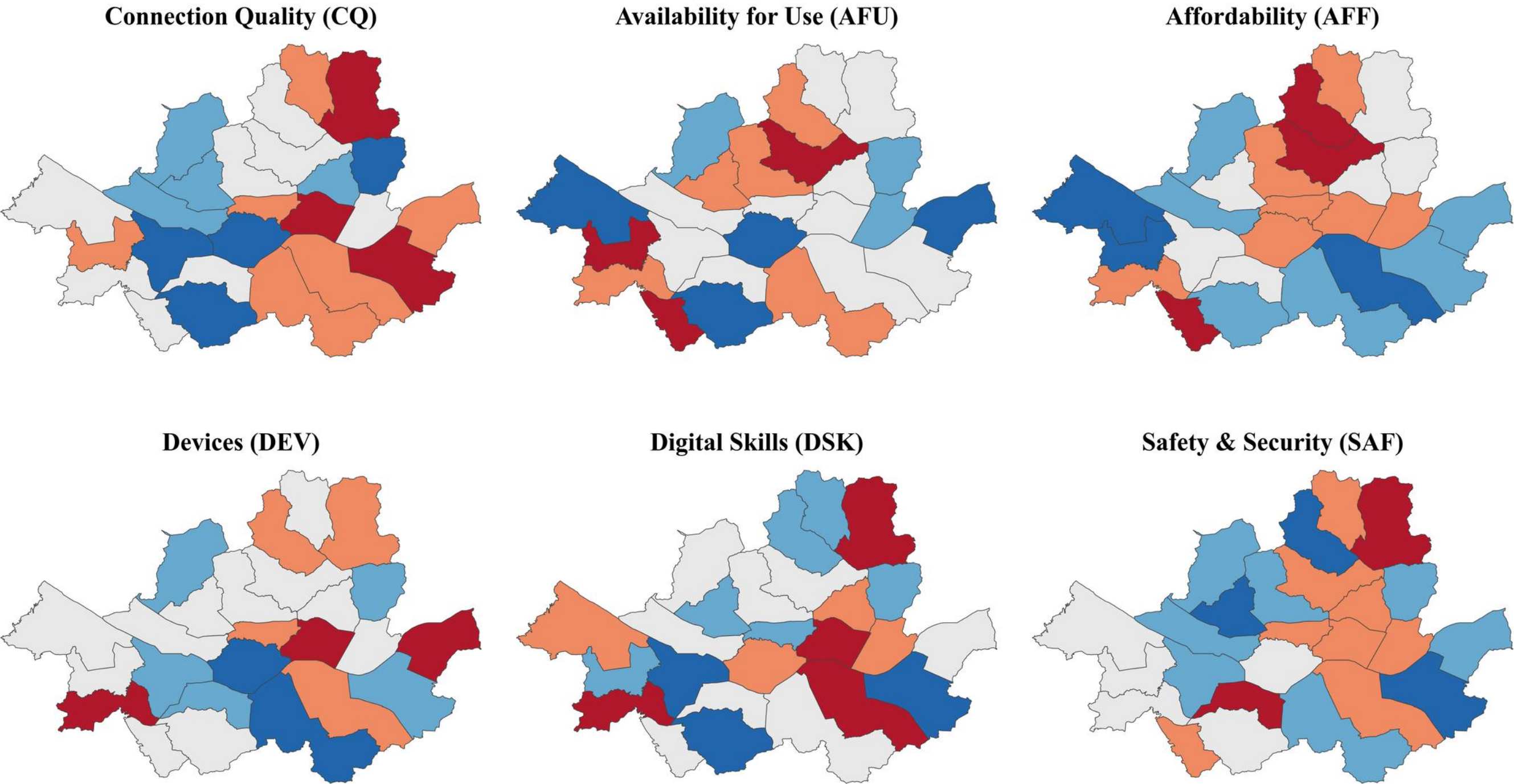
Cross-Checking the 3-1 Digital Deserts



Four Potential Digital Desert Districts Identified in Part 3-1

- Nowon-gu
 - Safety & Security -
 - Digital Skills -
 - Connectivity -
 - Jungnang-gu
 - Connectivity +
- Gangbuk-gu
 - Affordability +
- Dobong-gu

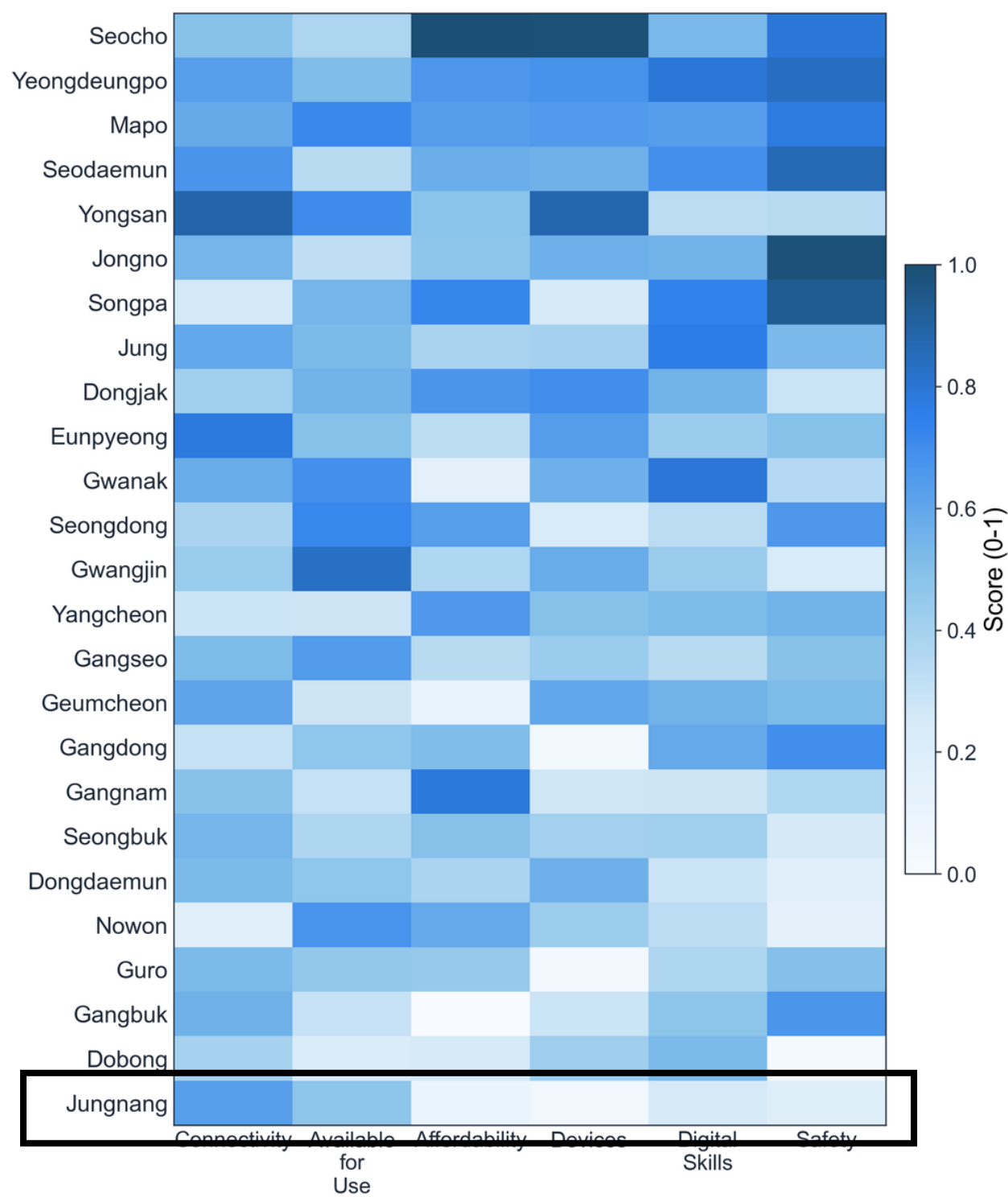
Where Divergence Clusters (z_shift map)



Interpretation Standardized divergence (z_shift) mapped across districts and dimensions; clusters flag where to look closer, not deprivation size.

Lived Event: Jungnang-gu Cable Fault

Connectivity +



According to the indicator, Jungnang-gu's Connectivity score lies between 0.4 and 0.6

SUPPORTED 2

Jungnang-gu Mangu 3-dong

Connection quality

2025-03-30 12:07 (Sun) KST · #730780200

Jungnang Dlive. Is your internet working?? Mine isn't :(And they're not answering the phone.....

Jungnang Dlive. Is your internet working?? Mine isn't :(And they're not answering the phone..

SUPPORTED 2

Jungnang-gu Muk 2-dong

Connection quality

2025-07-19 04:15 (Sat) KST · #731778513

Half-clear sky, then thunder, then a loud explosion sound! And now the internet is down.. Anyone else's SKT in...

SUPPORTED 2

Jungnang-gu Muk 2-dong

Connection quality

2025-07-19 13:25 (Sat) KST · #731780905

Is SK internet down around Muk-dong?? Both TV and internet have been down since morning—is it just us?...

SUPPORTED 2

Jungnang-gu Muk-dong

Connection quality

2025-07-19 19:22 (Sat) KST · #731784182

Muk-dong, still... the internet's down.. Funny thing—it worked fine this morning, then around 10 a.m. two repa...

Recurring problems in local cable TV and cable-based internet services continue to be reported.

Lived Event: Jungnang-gu (continued)

Connectivity +

SUPPORTED 2

Jungnang-gu Muk 2-dong

Connection quality

2025-07-19 04:15 (Sat) KST · #731778513

Half-clear sky, then thunder, then a loud explosion sound! And now the internet is down.. Anyone else's SKT in...

SUPPORTED 2

Jungnang-gu Muk 2-dong

Connection quality

2025-07-19 13:25 (Sat) KST · #731780905

Is SK internet down around Muk-dong?? Both TV and internet have been down since morning—is it just us?...

SUPPORTED 2

Jungnang-gu Muk-dong

Connection quality

2025-07-19 19:22 (Sat) KST · #731784182

Muk-dong, still... the internet's down.. Funny thing—it worked fine this morning, then around 10 a.m. two repa...

SUPPORTED 1

Jungnang-gu Muk-dong

Connection quality

2025-07-19 05:00 (Sat) KST · #731778572

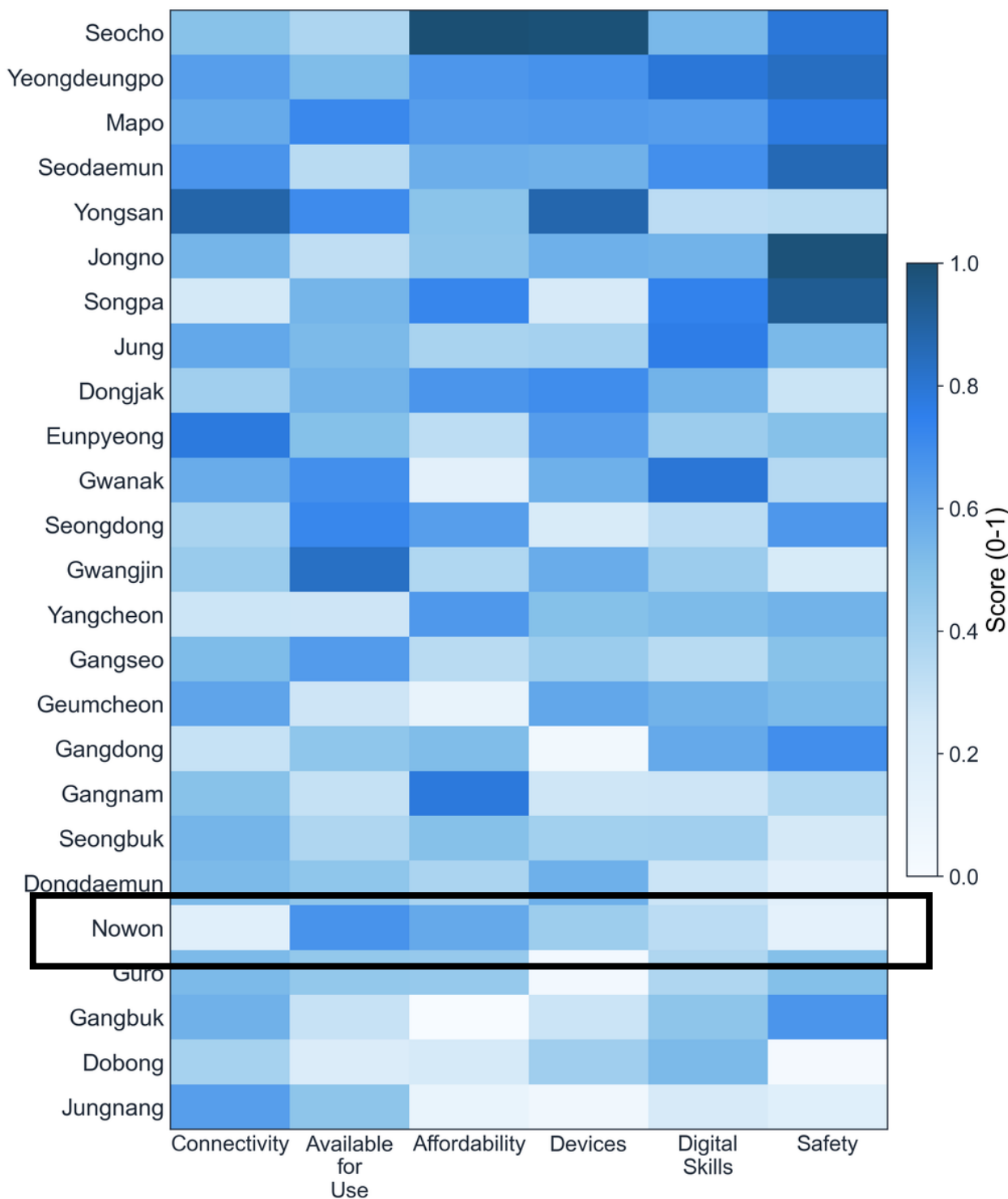
Power outage with thunder (check your breaker). Around 4:05 I realized the power went out along with a thunder...

Continued occurrence of weather-related cable communication disruptions — heavy rain, thunderstorms, and lightning.

Lived Event: Nowon-gu

CARRIES FORWARD All three analyses converge in Part 04: people-centered, place-sensitive, and a capability / safety / affordability bundle.

Nowon-gu, Safety, Digital Skills, and Connectivity were underrepresented relative to expectations.



SUPPORTED 1

Nowon-gu Sanggye 9-dong

Safety & security

2025-09-27 21:16 (Sat) KST · #732418409

[Nowon-gu] Notice on government system outage from the National Information Resources Service fire. [Web] Hell...

[Nowon-gu] Notice on government system outage from the National Information Resources Service fire. [Web] Hello, this is the Nowon-gu mayor. A fire yesterday (9/26) at the National Information Resources Service (Daejeon) has halted major online civil-service systems such as Gov24. Mobile ID and Gov24 identity verification are currently unavailable, so issuance of online certificates is partly restricted. Unavailable services: unmanned civil-service kiosks, mobile ID, Gov24, e-People, and other linked government systems. Available services: family relation records, resident registration, seal, passport, land/building ledgers, health certificates, basic-livelihood certificates, etc., issued in person at the district or community center with a physical ID (passport at the district office only; issuance delayed). Recovery-consumption coupon applications via card companies online proceed normally, but prepaid cards issued at community centers may be delayed.

- Identify potential district-level drivers of digital support capacity and assess their applicability to policy contexts in other regions
- Or this may imply that older adults in the district(21.4%) experience difficulties in digital usage.

04

Policy Recommendation

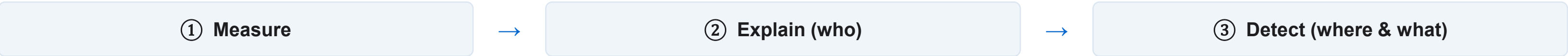
From findings to action

04

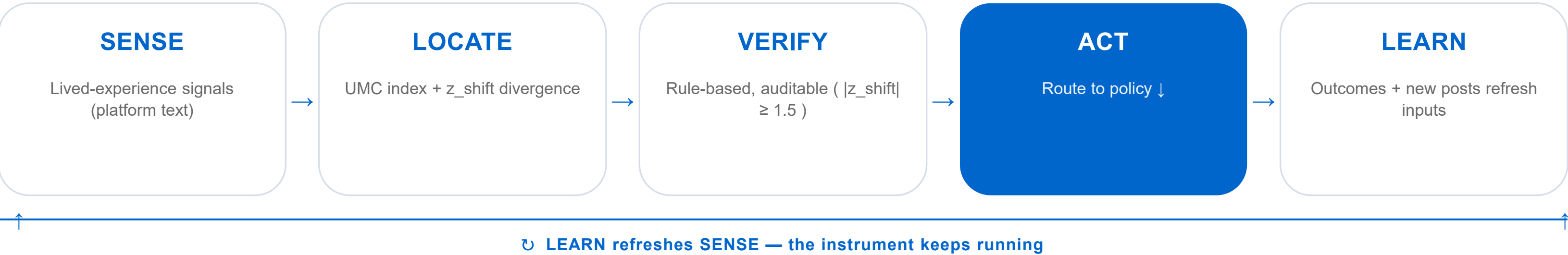
From a One-Pass Study to a Standing Instrument

Not a one-time diagnosis — a repeatable loop that feeds both people and place policy.

WHAT WE RAN ONCE



THE LOOP — PROPOSED OPERATING MODEL



FORK OUT OF ACT

PEOPLE · POLICY 1

A people-centered measure for the profile most consistently tied to lower digital use.

PLACE / MEASUREMENT · POLICY 2

A standing measurement layer that turns platform text into an auditable place signal.

Point-of-Need Digital Support for Low-Educated Older Adults

A demand-responsive model that delivers help at the moment a problem arises — pairing skills support with fraud and phishing protection.

WHY

Evidence

HLM identifies low education and older age as the largest, most consistent digital-use gap. Platform signals add that these needs — skills help and fraud/safety — surface at the moment of lived difficulty (e.g., "formatting is too hard, is there anyone nearby who can help?"), not on a fixed training schedule.

WHO

Target

Adults aged 65+ with middle-school education or below. District profiles (e.g., Nowon-gu, 21.4% elderly) are used only to prioritize outreach, not to define eligibility.

WHAT

Programme

- Shift from scheduled classroom literacy to point-of-need support:
- ✓ Neighborhood help desks and roving support embedded in care networks (senior welfare centers, gyeongrodang, Dolbom-SOS).
 - ✓ Just-in-time help when problems occur (device setup, civil services, payments), bundled with harm-prevention (phishing, fraud, authentication).
 - ✓ Active outreach to at-risk residents who do not self-request.

KEY TAKEAWAY Help the people most affected first, and close the gap between when a need arises and when help arrives — the gap our platform data uniquely measured.

A Resilience Gap the Index Misses — Weather-Driven Comms Outages

Lived-experience signals expose episodic connectivity failures that static indicators — and routine line-maintenance — do not capture, calling for a distinct disaster-resilience response.

WHY

Problem

Static infrastructure indicators capture steady-state connectivity but miss episodic, experiential failures. Jungnang-gu's connectivity index sits mid-range (0.4–0.6), yet residents repeatedly report weather-driven cable and internet outages (Muk-dong, Mangu-dong). The administrative picture reads "fine"; lived experience says otherwise.

WHAT

Recommendation

For districts the signal layer flags (e.g., Jungnang-gu), treat weather-driven communication-disruption resilience as a dedicated response category — separate from the line-maintenance track — and adopt the lived-experience signal as its early-warning input.

HOW

Use

- ✓ Periodic z_shift screening to surface candidate cells — an early-warning flag, not a prevalence estimate.
- ✓ Route flagged cells to the district for verification and a resilience response: redundancy / detour routing, backup power, recovery stockpiles, emergency comms.
- ✓ Feed verified findings back into the Policy 1 outreach targeting.

KEY INSIGHT Jungnang-gu leads the routine overhead-line maintenance track (aesthetics-and-safety, MSIT-evaluated). Weather-driven comms-disruption resilience redundancy, backup power, rapid recovery, emergency comms, real-time monitoring — is a different policy category that neither static UMC indicators nor a maintenance program covers. The lived-experience signal flags exactly this blind spot; the two domains overlap only at undergrounding.

Thank you

Q&A

Appendix

<https://github.com/ITU-project-team/part1>

<https://github.com/ITU-project-team/part2>

https://github.com/ITU-project-team/01_text_preprocessing

https://github.com/ITU-project-team/02_bayesian

https://github.com/ITU-project-team/03_inference

<https://github.com/ITU-project-team/daangn-crawler>